

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 3

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Topic

Classification of Admissible Field Equations

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Classification of Admissible Field Equations. State all assumptions and conventions.

$$K = \sum_{n=1}^{75} c_n k_n$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$K_{28}[\Phi] = \kappa \rho$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$\Delta\chi^2 = \sum_a (R_a^{new} - R_a^{old})^2$$

Show enough intermediate work that another student can reproduce the calculation.